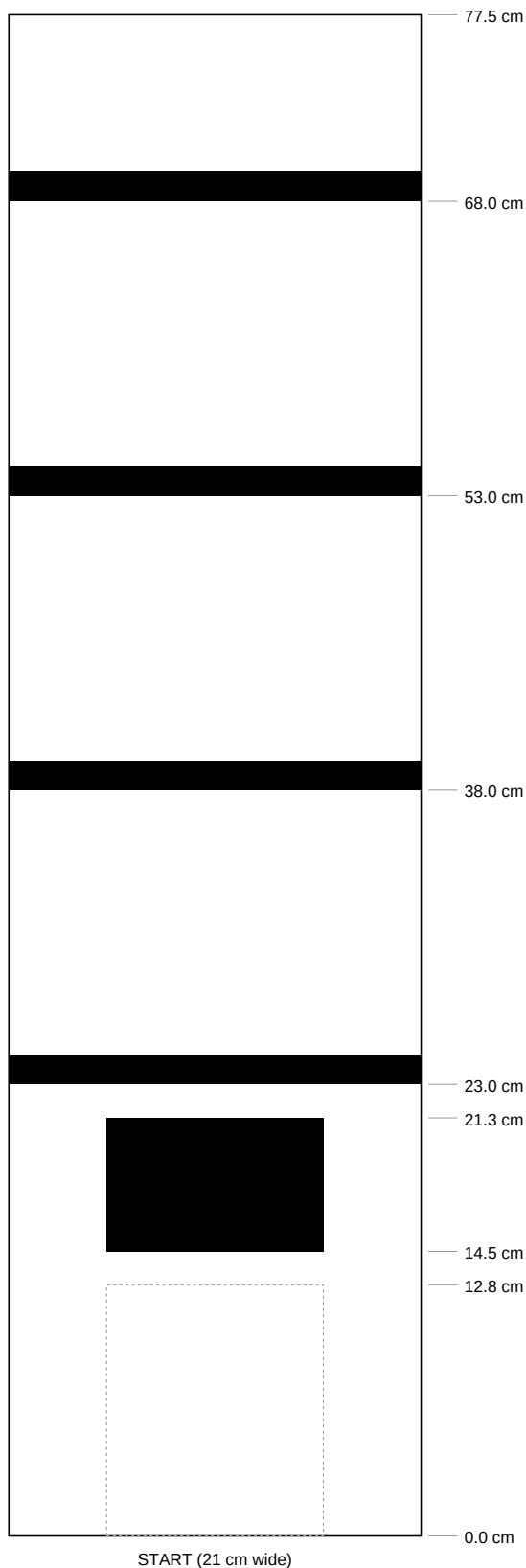


Calibration arena 21 x 77.5 cm

Print on 5 landscape A4 sheets and join them with tape.



1. Print settings

LANDSCAPE orientation, 100% scale.

Turn off "fit to page" / "shrink oversized pages".

Check the ruler printed on every sheet: it must measure exactly 100 mm.

2. Preparation

Trim each sheet along the solid line at the bottom.

Do not trim the top edge: that 1.5 cm flap is the overlap.

The vertical dashed lines mark the corridor edges (21 cm): trim there if you need the exact width.

3. Assembly

Start from sheet 1 (START) and work upwards in order.

Lay the trimmed edge of sheet n+1 onto the horizontal dashed line of sheet n, matching the registration crosses on both sides.

Tape from the back; on the front use matte tape only, to avoid specular reflections.

Black areas stop at the fold line on purpose: the flap is hidden by the next sheet, which carries the rest.

4. Walls

Only two walls are needed, marked by the "PLACE WALL HERE" labels: the long RIGHT side, and the one across the START line. The left side and the 77.5 cm end are left open.

Walls sit ON the line, outside the floor, so the clear inner width stays exactly 21 cm.

They must be WHITE and MATTE, like the floor, so they do not alter the readings of the reflective sensors.

5. Robot start position

The dashed outline on sheet 1 is the robot footprint.

Before every run, push the robot BACKWARDS until its rear touches the START wall, so the rear edge sits on the START line, and centre it inside the outline.

That contact, not the outline, sets the zero of the run.

6. Note on the black areas

Laser black is often glossy. Alternatively print the outlines only and fill them with matte black electrical tape, which is more uniform for reflective sensors.

Dimensions used

width 21 cm - total length 77.5 cm

robot footprint: 12.8 cm long, 11.0 cm wide, rear at START

narrow patch: 6.8 cm thick, 11.0 cm wide, from 14.5 to 21.3 cm

START -> first full-width band: 23 cm

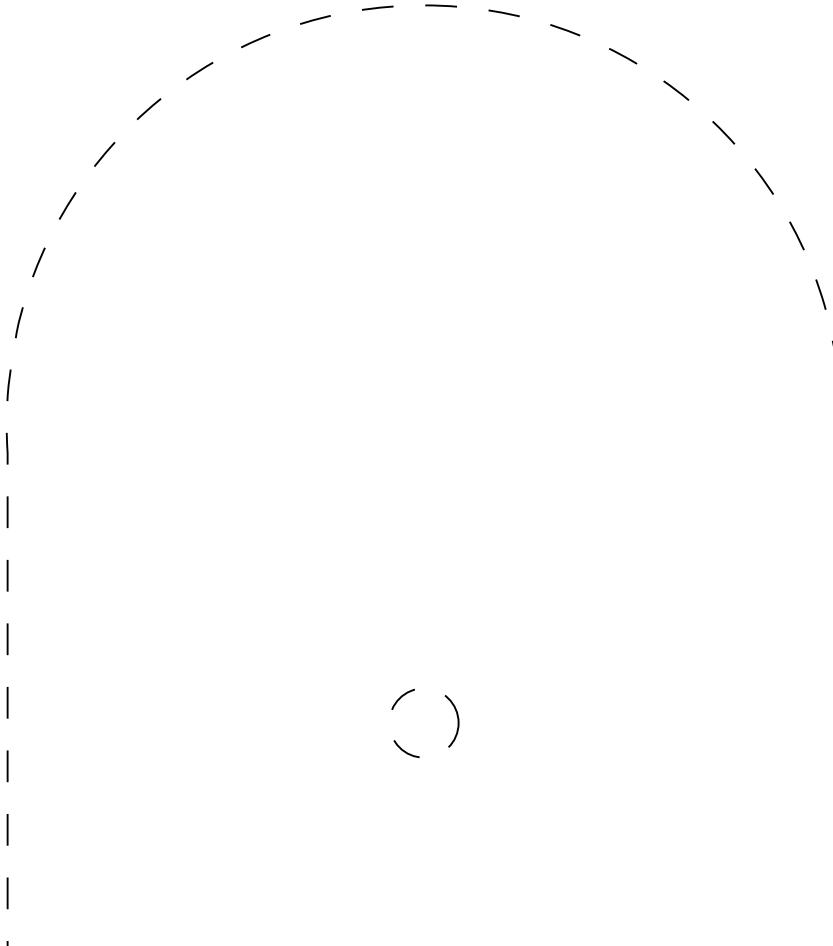
black bands: 1.5 cm thick, 15 cm pitch (edge to edge), 4 of them

last band ends 8.0 cm before the END line

<<< towards START

exactly 100 mm = correct scale

overlap flap: lay sheet 2 on top of this area



PLACE RIGHT WALL HERE

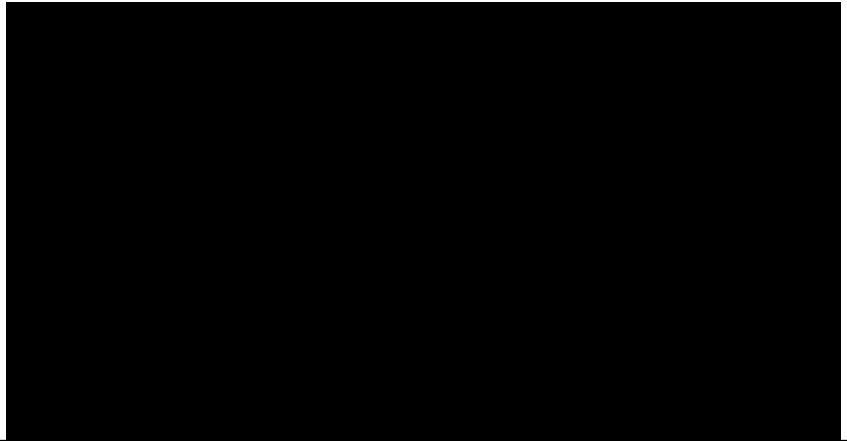
START (0 cm) --- PLACE WALL HERE

<<< towards START

exactly 100 mm = correct scale



overlap flap: lay sheet 3 on top of this area

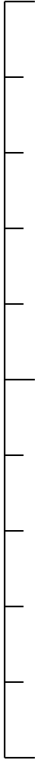


PLACE RIGHT WALL HERE

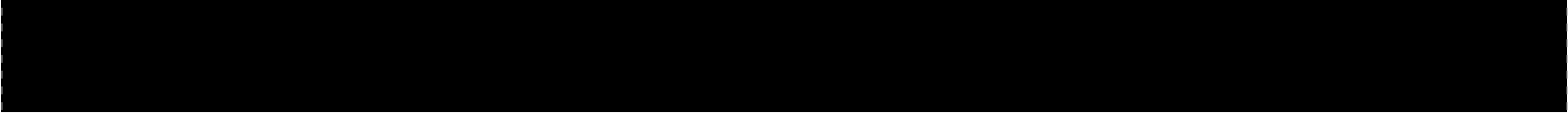
CUT HERE --- edge at 15.5 cm from START

<<< towards START

exactly 100 mm = correct scale



overlap flap: lay sheet 4 on top of this area



PLACE RIGHT WALL HERE

CUT HERE --- edge at 31.0 cm from START

<<< towards START

exactly 100 mm = correct scale



overlap flap: lay sheet 5 on top of this area



PLACE RIGHT WALL HERE

CUT HERE --- edge at 46.5 cm from START

<<< towards START

exactly 100 mm = correct scale

